

Cetaceans

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Cetacea
FAMILIES	14
SPECIES	89

CLASSIFICATION NOTE

This book adopts the traditional method of classifying cetaceans, whereby the order is divided into 2 suborders: baleen whales and toothed whales. It is not certain how they are related to one another, but genetic and morphological evidence suggests the order Cetacea is most closely related to the family Hippotamidae, which is in the order Artiodactyla.

Baleen whales
see pp.246–9
Toothed whales
see pp.250–261

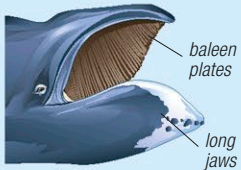
Cetaceans—whales, dolphins, and porpoises— are perhaps the most specialized of all mammals, with their fish-shaped, hairless body; a flipperlike front limbs; and vestigial hind limbs (located within the body wall). However, they are true mammals: they breathe air with lungs, and they have mammary glands with which they suckle their young. Cetaceans, which can be divided into baleen whales (such as the humpback whale and the right whale), and toothed whales (such as dolphins and porpoises), are found throughout the world’s seas, and some species live in tropical and subtropical rivers. Many species, including the blue whale, have been hunted in such numbers that they are in danger of extinction.



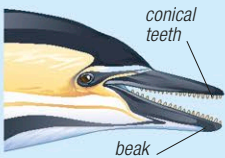
MAMMALS

Anatomy

Cetaceans have a hairless, streamlined body to reduce water turbulence. External projections are reduced to the essentials: flippers for steering, a tail with 2 boneless, horizontal flukes (fish have vertical flukes), and usually a dorsal fin for stability. Even the genitals are concealed within folds. Other adaptations to underwater life include a thick layer of blubber (fat and oil) beneath the skin, which conserves body heat, and light, spongy, oil-filled bones. Cetaceans breathe through one (toothed whales) or 2 (baleen whales) blowholes—muscular nostrils usually situated on the top of the head. Toothed whales have a brain that is relatively as large as that of primates, and they are known for their intelligence. Baleen whales have a relatively smaller brain.



BALEEN WHALE
Instead of teeth, a baleen whale has 130–400 horny (baleen) plates on each side of the upper jaw. The inner edge of each plate has bristles used to sieve food.



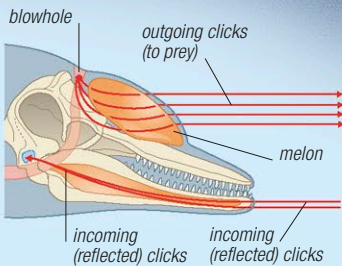
TOOTHED WHALE
The teeth of the toothed whale group are simple, conical, and in most species numerous. The jaws may be extended to form a beak, as seen in dolphins.



POWERFUL TAIL
In cetaceans, the main propulsive force is the up-and-down movement of the tail. This motion is powerful enough for a whale, such as this humpback, to push two-thirds of its body out of the water when breaching.

Senses

Cetaceans have extraordinarily sensitive hearing. Vision is reasonable—underwater it is excellent up to about 3¼ft (1 m), and in air up to 8¼ft (2.5 m), but color vision is very limited. Some species can focus both eyes ahead, above, or behind them, and some can move their eyes independently. Some freshwater dolphins, however, are nearly or entirely blind. Members of the toothed whale group produce high-frequency clicks for echolocation and can also communicate using a wide range of sounds audible to humans. Other cetaceans employ a variety of vocalizations, but these are not as well studied. Cetaceans have no sense of smell.



USING ECHOLOCATION
Toothed whales avoid obstacles and catch prey by emitting high-frequency clicks that are reflected off objects in their path. The melon (a fluid-filled swelling) changes shape to focus the clicks. Incoming clicks pass through the jaw.

BREACHING



THE ASCENT
As this southern right whale begins to breach, it surfaces vertically, creating little water turbulence.

THE FLIPPERS APPEAR
The front flippers clear the surface as the whale continues to propel itself upward.

STARTING TO GO DOWN
As the upward surge, which is powered by the tail, is completed, nearly half of the whale is exposed.

ASCENT COMPLETED
As the whale crashes back into the water, it also turns onto its side.